Project-Utopia: A Tool-Grounded Multi-Channel Benchmark for LLM Hierarchical Decision-Making, Stale-Coherence Memory Failures, and Cross-Vendor Channel Routing

Anonymous Authors[1]

[1]Affiliation withheld for double-blind review

2026-05-10 — Submission draft

Abstract

Existing LLM-agent benchmarks evaluate single agents picking single tools [24, 30, 33, 35, 36, 55] or flat MARL policies emitting primitive actions in stateless games [1, 12, 31], leaving the regime where an LLM agent **orchestrates a fixed, schema-validated tool surface across multiple parallel decision channels** largely under-evaluated — despite this being the canonical pattern in production agent systems [3, 29, 37, 45]. We introduce **Project-Utopia**, a tool-grounded benchmark exposing four hierarchical LLM decision channels (*environment-director*, *npc-policy*, *strategic-plan*, *colony-agent*) that orchestrate a deterministic, schema-validated tool surface (path queries over a versioned grid, group steering, FSM intent application, build planning). Schema validation and numerical guardrails ensure **three-tier reproducibility**: (i) bit-identical fallback mode (verified at 7 200 ticks), (ii) hardware-independent LLM-on mode via LayerCast [53], (iii) statistically stationary production-LLM mode.

We propose a four-layer metric stack combining per-tick dimension plugins (Resource-Allocation Efficiency, Group Dynamics, Memory Degradation, Decision Token Efficiency, Hierarchical Coordination) with MeltingPot-style sandwich normalization [1], Crafter geometric mean [12], HELM Mean Win Rate [22], and Pluribus-style PolicyValue Baseline [6] for $\sim$2–3$\times$ seed-variance reduction. Statistical inference follows Bowyer et al. [5]'s ICML 2025 Spotlight recommendation of Beta-Binomial credible intervals for small-N.

Across nine experiments evaluating seven LLM families on five scenarios with ten seeds each ($\sim$2 250 runs total), we demonstrate **(i)** hierarchical decomposition outperforms a matched-budget flat baseline by $\sim$0.10 RAE (matching ChatDev's -44% role-removal effect size [34]); **(ii) cross-vendor channel-axis** heterogeneous LLM mixes (Anthropic + OpenAI + open-weight) achieve near-strong-on-all RAE at $\sim 1/3$ the cost — going beyond X-MAS [27]'s per-task routing and Anthropic's intra-family mix [3]; **(iii)** the first **three-axis joint memory analysis** (Recall(t) $\times$ Drift(t) $\times$ Performance(t)) on continuous tick-level naturally-accumulated context, dissociating verbal recall from action-grounded recall in ways LongMemEval [49] and MemoryArena [13] cannot.

Project-Utopia is released as a self-contained Python package (`pip install project-utopia`, 621 tests passing, $\sim$5 500 LOC) with one-line CLI reproducibility, a record-replay LLM cache via pydantic schemas, and full JSON-NDJSON paper-run pipeline. Code, prompts, and 7 200-tick determinism manifests are at [ANONYMIZED-REPO-URL] (de-anonymized for camera-ready).

1 Introduction

Production LLM agents rarely fail at single tool-call accuracy: AgentBench [24], GAIA [30], Toolformer [36], Gorilla [33], and ToolLLM [35] all establish that current frontier models pick and parameterize individual tools competently. The mode that actually breaks deployments is *coherent multi-channel tool orchestration under stale long-horizon context*: an agent that emits a structurally valid build directive, a structurally valid group-steering directive, and a structurally valid environment update — each correct in isolation, jointly inconsistent because some channel still references a directive evicted from working memory hours ago.

This pattern — one agent, several parallel decision channels, a fixed schema-validated tool surface — is canonical in production systems [3, 14, 29, 34, 37], in robotics [4, 45], and in ReAct-style [52] reasoning loops scaled up to multi-channel orchestration. Yet existing LLM-agent benchmarks evaluate a substantially narrower slice of this design space:

- **Single-agent tool-use & tool-selection benchmarks** (AgentBench [24], GAIA [30], WebArena [55], AgentBoard [26], TheAgentCompany [51], Toolformer [36], Gorilla [33], ToolLLM [35], MINT [48]) measure success on episodic tasks with single-LLM dispatch over an open or learned tool set, with no parallel decision channels.
- **Flat MARL benchmarks** (MeltingPot [1], Crafter [12], BALROG [31]) test policies that emit primitive actions, with no natural-language directive surface.
- **Multi-agent LLM frameworks** (ChatDev [34], MetaGPT [14], AgentVerse [7], X-MAS [27], HuggingGPT [37]) explore agent collaboration but seldom combine *cross-vendor heterogeneity* with *long-horizon* orchestration, structured *schema-validated* tool-grounded directives, and a *deterministic tool surface*.
- **Long-context / memory benchmarks** (LongMemEval [49], RULER [15], MemoryArena [13], MemoryAgentBench [16]) probe verbal recall, but rarely couple recall failures to downstream *task* signal.

This gap matters because the most common production failure mode is not "the LLM emitted an invalid action" — schema validators catch that — it is *coherent but stale* decisions: the strategic-plan channel still references a long-evicted directive, the npc-policy channel allocates resources to a problem solved an hour earlier, the colony-agent channel commits to a build that contradicts the environment-director's just-issued threat update. Detecting this failure mode requires (a) parallel decision channels running at distinct cadences, (b) a long horizon where directives cumulate and decay naturally rather than being injected as synthetic distractors, and (c) a deterministic tool surface so that hardware/timing variance does not mask reasoning failures.

Contributions. We introduce **Project-Utopia**, a tool-grounded multi-channel LLM-agent benchmark that closes this gap with four contributions:

C1 — Action-Space Contract. A 4-channel LLM action surface (environment-director, npc-policy, strategic-plan, colony-agent) emitting schema-validated, tool-grounded directives. The agent has *no syscall-level tool choice*; instead, each channel's directive is deterministically grounded by a fixed, queryable tool (path query over a versioned grid, group-steering policy, FSM intent applier, build planner). Schema validation and numerical guardrails ensure **three-tier reproducibility**: (i) bit-identical fallback mode (verified at 30, 1 800, and 7 200 ticks; hashes `e006ea96...`, `e99d4fcb...`, `87ecd82b...`), (ii) hardware-independent LLM-on mode via LayerCast [53], (iii) stationary distribution production-LLM mode.

C2 — Methodological. A four-layer metric stack: (1) per-tick dimension plugins emit raw signal; (2) a dimension normalizer applies six transform types (identity, invert, reciprocal, clip-scale, symmetric-rescale, exponential-decay) to project all 19 score keys into $[0, 1]$ benefit form; (3) MeltingPot-style sandwich normalization [1] computes $(\text{LLM} - \text{fallback})/(\text{oracle} - \text{fallback})$ on aligned per-seed paired runs; (4) Crafter geometric mean composite [12] for resource-allocation, HELM Mean Win Rate [22] for cross-model leaderboards. Variance reduction follows Pluribus AIVAT [6]; statistical inference follows Bowyer et al. [5].

C3 — Empirical. Nine experiments yielding three findings: (a) hierarchical decomposition advantage at matched token budget; (b) *cross-vendor channel-axis heterogeneous mix* achieves near-strong-on-all RAE at $\sim 1/3$ the cost — going beyond X-MAS [27] per-task routing and Anthropic [3] intra-family mix; (c) the first *three-axis joint memory analysis* (Recall × Drift × Performance) on continuous tick-level naturally-accumulated context, dissociating verbal recall from action-grounded recall.

C4 — Position-defining. First benchmark in the 2025–2026 LLM-driven sim landscape combining (i) a deterministic 96×72 tile economy, (ii) 4-channel parallel commands per tick, (iii) FSM-worker emergent multi-resource economy, (iv) seeded year-scale reproducibility, and (v) cross-vendor channel-axis heterogeneous LLM mix evaluation. Each ingredient has prior art individually; the 5-way combination has none.

Differentiation from prior tool-use frameworks. **ReAct** [52] interleaves reasoning and acting for a *single* agent emitting free-form action strings; we keep the reasoning–acting loop but split it across four parallel channels with schema-validated directives. **Toolformer** [36] learns when to invoke a small set of tools from self-supervision; we fix the tool set a priori and evaluate orchestration quality rather than tool selection. **Gorilla** [33] and **ToolLLM** [35] score API-call accuracy on 1 000s of REST endpoints; we score long-horizon outcomes downstream of a fixed 4-tool surface, so the failure mode of interest is *coherence across channels* rather than picking the right endpoint. **MINT** [48] evaluates multi-turn tool use with user feedback; we evaluate multi-channel tool orchestration with no human-in-the-loop and per-tick deterministic grounding. **HuggingGPT** [37] dispatches tasks to model-zoo tools through a planner LLM; like us, it routes per task, but it has no parallel-channel cadence and no long-horizon memory axis. **Voyager** [46] discovers skills in Minecraft through self-generated code; we keep the tool surface fixed and measure orchestration under stale context rather than skill discovery.

Relation to concurrent work. Three concurrent benchmarks deserve specific differentiation. **Orak** [19] evaluates LLMs on 12 video games via an MCP plug-and-play interface; we differ by going *depth-first* on a single deep simulator with 4-channel parallel commands and dimension-decomposed scoring instead of *breadth-first* across shallow MCP shells. **X-MAS** [27] demonstrates heterogeneous-LLM benefits on per-task routing across 27 LLMs; we route *per channel within a single long-horizon survival task* and add cross-vendor mixes that X-MAS does not test. **MemoryArena** [13] probes action-grounded recall in session-discrete agent loops; we run on *tick-continuous, naturally-accumulated* context and report a three-axis decomposition that joint-analyzes recall against drift and task performance.

Roadmap. Section 2 surveys related work. Section 3 describes the tool surface and 4-channel decision contract. Section 4 presents the four-layer metric stack. Section 5 reports the nine experiments. Section 6 discusses cross-experiment findings, Section 7 states limitations and future work, and Section 8 describes the reproducibility package.

2 Related Work

We organize related work along six axes, mapping each to the contribution claims (C1–C4) of §1. A complete prior-art table for the 75+ benchmarks reviewed appears in Appendix **??**.

2.1 LLM Tool-Use and Tool-Augmented Agents

The closest immediate neighbours of Project-Utopia are agent frameworks that pair an LLM with a tool surface. **ReAct** [52] formalized interleaved reasoning and acting for a single agent; subsequent work scaled the action side: **Toolformer** [36] taught models to self-invoke a small tool inventory; **Gorilla** [33] and **ToolLLM** [35] pushed the inventory to 10^3–10^4 real APIs and scored selection accuracy; **MINT** [48] added multi-turn user feedback; **HuggingGPT** [37] used an LLM planner to dispatch sub-tasks to model-zoo tools. These works converge on a shared methodology: *measure single-channel selection or single-turn invocation accuracy*. Project-Utopia is complementary: it fixes the tool surface (4 tools $\times$ 4 channels) and measures *cross-channel orchestration coherence under stale long-horizon context*. **Reflexion** [38] adds verbal self-improvement to ReAct loops; our colony-agent channel implements an analogous reflect step but uses simulator-objective feedback rather than the agent's own critique.

2.2 Single-Agent Tool-Use and Web Benchmarks

AgentBench [24] (N=8 environments), GAIA [30] (N=466 tasks), WebArena [55] (N=812 web tasks), AgentBoard [26] (Progress Rate scoring), BALROG [31] (N=6 RL games), and recent TheAgentCompany [51], OdysseyBench [47], HeroBench [2] all evaluate *single* agents on *episodic* tasks. We share their commitment to programmatic state-based oracles (rather than LLM-as-judge); we differ in dispatching *four* parallel decision channels over a fixed schema-validated tool surface at distinct cadences within a single continuous task.

2.3 LLM-in-Game Benchmarks

Voyager [46] (Minecraft skill discovery), Cicero [29] (Diplomacy negotiation), Generative Agents [32] (Smallville believability), SmartPlay [50] (N=6 games $\times$ 9 capabilities), and MineDojo [10] (N=3 000+ tasks) established LLM-in-game evaluation. The most direct concurrent competitor is **Orak** [19], which evaluates LLMs on 12 video games (including Stardew Valley and StarCraft II) through MCP plug-and-play. Project-Utopia differs by going *depth-first* on one deep colony simulation with a 96$\times$72 deterministic tile economy, FSM workers, and a multi-resource processing chain — producing per-channel and per-dimension scores that Orak's flat per-game scalars cannot.

2.4 Memory and Long-Context Benchmarks

Long-context evaluation has rapidly matured in 2024–2026: RULER [15] (effective context length, 13 task families), LongMemEval [49] (long-term chat memory), Lost-in-the-Middle [23] (positional bias U-shape), InfiniteBench [54] ($\geq$ 100k tokens), and the most recent MemoryArena [13], MemoryAgentBench [16], Evo-Memory [9], Memora [28], and HELM Long Context [42]. These probe *verbal* recall on synthetic distractors; only MemoryArena couples recall to action grounding, but in *session-discrete* agent loops. Project-Utopia's contribution is the first three-axis joint analysis (Recall $\times$ Drift $\times$ Performance) on *tick-continuous, naturally-accumulated* context, capturing a dissociation between verbal-but-not-action recall that prior single-axis work cannot. The Illusion-of-Diminishing-Returns analysis [40] provides the theoretical underpinning for our drift curves.

2.5 Multi-Agent LLM Coordination

ChatDev [34] (5 software-engineering roles in chat-chain), MetaGPT [14] (publish-subscribe message pool), AgentVerse [7] (4-stage recruitment), CAMEL [21] (2-agent role-playing), and Concordia [44] (Game-Master broker) established multi-LLM coordination patterns. Crucially, **X-MAS** [27] (May 2025) is the first to *systematically* demonstrate heterogeneous-LLM advantages on per-task routing across 27 LLMs × 5 domains × 5 functions, and **Anthropic's Multi-Agent Research System** [3] validated intra-family Opus + Sonnet mixes in production. Subsequent work clarifies that naive multi-agent debate often *degrades* performance [43], and that heterogeneity is the antidote. Project-Utopia's E3 experiment (§5.4) extends this literature by adding three orthogonal axes: *cross-vendor* mixes (Anthropic + OpenAI + open-weight, vs. X-MAS's mostly single-vendor scoring), *per-channel* routing (vs. per-task), and *long-horizon survival* (vs. per-task isolated runs).

2.6 RL Evaluation Methodology (Borrowed)

Although Project-Utopia is not an RL benchmark, the metric stack borrows four ingredients from RL game-playing AI methodology: Crafter [12]'s geometric-mean composite (punishes single-axis collapse); MeltingPot 2.0 [1]'s sandwich normalization (focal − random)/(exploiter − random); Pluribus AIVAT [6]'s PolicyValue Baseline (2–3× variance reduction, §5.9); and TrueSkill [4] / α-rank [20] for low-sample multi-LLM cells. AlphaStar [45] and FTW Capture-the-Flag [17] inspire our matched-budget fairness convention.

2.7 Reproducibility and Methodology

Bowyer et al. [5] (ICML 2025 Spotlight) demonstrated that the central-limit-theorem normal approximation under-covers at LLM-benchmark sample sizes ($N < 300$) and recommended Beta-Binomial credible intervals as the community default; we adopt this throughout. Yuan et al. [53] (NeurIPS 2025 Oral) showed that bf16 LLM inference can swing accuracy by 9 percentage points across hardware, and proposed LayerCast for hardware-independent bit-identical inference — enabling our Tier 2 reproducibility claim. Cheng et al. [8] (NeurIPS 2025 Position Paper) argued *against* agent-as-judge for high-stakes capability evaluation; following their guidance, all Project-Utopia metrics are simulator-objective, never LLM-judged. Siddiq [39] introduced the Reproducibility Maturity Model (RMM) with seven smell categories; Project-Utopia targets RMM Tier 4+ (§8). Liu et al. [25] formalized cost-aware Pareto reporting; we adopt $\alpha \cdot \text{tokens} + \beta \cdot \text{tool\_calls}$ following BATS. The MathArena [41] streaming-benchmark template inspires our scenario rotation (§7).

3 Project-Utopia Architecture

3.1 Overview: A Tool-Grounded 4-Channel Action Surface

Project-Utopia exposes a 4-channel LLM action surface above a fixed, queryable tool layer. The agent has *no syscall-level tool choice*: it emits structured directives, and the tool layer deterministically grounds them to per-entity actions via fixed tool implementations. This is the language convention of ReAct [52] and Toolformer [36] extended to a multi-channel setting: each channel binds to one tool, and the schema is the agent–tool interface contract.

The tool layer offers four deterministic, queryable implementations that LLM channels indirectly invoke through directives: **(T1) path query tool** — A* over a versioned grid, cached by $(\text{faction}, \text{gridVersion}, \text{costVersion})$; **(T2) group steering tool** — Boids with per-group profiles applying `intentWeights` to NPC steering; **(T3) FSM intent application tool** — per-NPC finite-state machine driven by `POLICY_INTENT_TO_STATE` mapping; **(T4) build planner tool** — PlanExecutor over BuildSystem, grounding multi-step plans tile-by-tile. A seeded RNG and a fixed system order tie these tools into a reproducible per-tick pipeline.

Concretely, the per-tick pipeline runs at $\Delta t = 1/30\,\text{s}$ through 17 systems (perception → LLM gates → policy applier → FSM → A* → Boids → integration → telemetry). The simulation hosts 20–50 NPC workers harvesting resources, building, and defending against periodic raids over 60–240 sim seconds (configurable up to 24 hours). The LLM is consulted at three nested

[Figure 1 placeholder] 4-channel LLM action surface bound to a deterministic 4-tool layer; tick pipeline; schema/guardrail boundary at `entity.desiredVel`. See `docs/ai-research/refactor-plan.md` §1.3 for the ASCII version this figure replaces.

Figure 1: Project-Utopia architecture: four LLM decision channels (*environment-director*, *npc-policy*, *strategic-plan*, *colony-agent*) emit schema-validated directives that deterministically ground onto a fixed 4-tool surface (path query, group steering, FSM intent applier, build planner). Schema validation and numerical guardrails ensure that LLM directives never violate hard physical or accounting invariants.

cadences: strategic every 90 s, environment / npc-policy every 5–15 s, colony-agent on event triggers.

Crucially, *the LLM never controls individual entities*. It produces low-rank, schema-validated, clamped *directives* that flow through the deterministic tool layer: a 4-channel directive stream → group policy applier (T3) → per-NPC FSM → A* target (T1) → Boids `desiredVel` (T2) → integrated position. This handoff is the deterministic boundary that makes bit-identical reproducibility possible at every channel even when the LLM itself is non-deterministic (see §8).

Figure 1 illustrates the architecture.

3.2 The Four LLM Decision Channels

Environment-director (*cadence*: 5–15 s, throttled by `tuning.environmentDecisionIntervalSec`). Emits directives — weather (enum: `clear|rain|storm`), `durationSec` (clamped 8–180 s), `factionTension` ($\in$ [0, 1]), `eventSpawns` (at most 3, each with intensity ≥ 0.4 and clamped duration) — that the *event scheduler tool* grounds into naturalistic stressors (storms, faction buildups, raid caravans). Functions as the environmental state-perturbation tool to which the policy channel must adapt.

NPC-policy (*cadence*: 5–15 s, additionally gated by threat ≤ 14 tiles, guard deficit, or active worker strikes). Emits per-group policy directives $\{\text{groupId}, \text{intentWeights} \in [0,3]^{|\mathcal{I}_g|}, \text{riskTolerance} \in [0,1], \text{targetPriorities} \in [0,3]^{|\mathcal{T}_g|}, \text{ttlSec} \in [8, 24\,\text{h}]\}$ that the *group steering (T2) + FSM intent applier (T3) tools* then ground: intent weights flow through `POLICY_INTENT_TO_STATE` into per-NPC FSM state preferences, while target priorities are interpreted by the A* path query tool (T1).

Strategic-plan (*cadence*: 90 s heartbeat plus crisis triggers — workers → 0, food/wood ≤ 5, threat ≥ 85 — with 15 s cooldown). Emits free-text colony-level objectives plus structured `primaryGoal` / `constraints` / `resourceBudget` directives that the *plan-arbitration tool* grounds by adjusting downstream channel budgets. Strategic plans are advisory; lower channels can override under raid posture clamps.

Colony-agent (*cadence*: sim-time gated, additionally triggered by plan stalls). Emits multi-step plan directives [(action, args, depends_on)] that the *build planner tool (T4)* (`PlanExecutor` over `BuildSystem`) grounds tile-by-tile. Wraps a Perceive → Plan → Ground → Execute → Evaluate → Reflect loop [38, 52], where Ground is the tool-invocation step.

3.3 Schema Validation and Guardrails: the Agent↔Tool Contract

Schemas and guardrails are not safety bolt-ons; they *are* the agent–tool interface contract — the same role as the typed JSON signatures of ToolLLM [35] and the function-call schemas of Gorilla [33], generalized to four parallel channels. All four channels share two filters before directives reach the tool layer. **ResponseSchema** performs JSON-Schema validation: unknown enum values are dropped, NaN values are rejected, field types are coerced. **Guardrails** clamps numeric ranges (weights $\in [0, 3]$, risk $\in [0, 1]$, TTL $\in [8\,\mathrm{h}, 24\,\mathrm{h}]$) and applies *raid-posture clamps*: when threat is imminent, intent weights for combat-relevant intents are forcibly raised regardless of LLM output. This contract makes the action space precisely $\sim 10^2$–10^3 continuous axes per directive, decoupled from entity count, and it ensures the tool layer never enters an invalid state even when the LLM emits adversarial or malformed JSON.

3.4 Determinism and the AgentAdapter Seam

The tool layer is deterministic given a seed. RNG flows through `project_utopia.app.rng`; A* paths are cached by (faction, gridVersion, costVersion); Boids accumulators run in fixed order; `PathWorkerPool` is disabled in `deterministic=True` mode. Verifying determinism empirically is non-trivial: we built `project_utopia.tools.audit.determinism_check` which runs the harness twice with the same seed and compares SHA-256 hashes of slim state snapshots. Three tiers are gated: 30 ticks, 1 800 ticks, and 7 200 ticks (240 sim-seconds). All three pass with bit-identical hashes (`e006ea96...`, `e99d4fcb...`, `87ecd82b...`; see Appendix A).

The **AgentAdapter** is the interface that any LLM (or fallback policy, or scripted oracle) must implement to drive the four channels:

Listing 1: AgentAdapter contract

```
async def request(self, channel: str, payload: Any,
                  options: dict | None = None) -> DecisionResponse:
    # channel in {environment-director, npc-policy,
    #             strategic-plan, colony-agent}
    # returns DecisionResponse(data, fallback, usage,
    #                          latency_ms, model, error, debug)
    # MUST NOT raise; on failure return a fallback DecisionResponse
    ...
```

Concrete adapters in our suite include: (1) `LLMClient` (OpenAI-compatible HTTP via litellm); (2) `NoopAgentAdapter` / `ScriptedOraclePolicy` (used for the sandwich-norm fallback and oracle terms); (3) `HTTPAgentClient` dispatching to a 9-cell cross-vendor routing config (`project_utopia.cli.agent_routing`); (4) `LayerCastAdapter` wrapping any inner adapter with deterministic-inference headers; (5) `RecordReplayCache` (VCR-cassette-style) for offline re-execution from a previous run's transcript.

`AdapterToLLMClient` bridges any `AgentAdapter` into the existing simulation systems by exposing the four legacy method signatures (`request_environment`, `request_policies`, `request_strategic`, `request_plan`) that the existing strategic / environment / brain / colony-director systems call. This single seam is what makes E3's 9-cell cross-vendor matrix possible without modifying any simulation system.

3.5 Tool-Interface Contract

Table 1 tabulates the four channels, their directive schemas, and the deterministic tool implementation that grounds each directive into per-tick state changes. This is the *action-space contract* of contribution C1: the agent's surface is exactly four typed channels into four named tools, and nothing else is reachable.

4 Metric Stack

Project-Utopia reports scores at four layers, each grounded in prior art (Figure 2).

Table 1: Channel → directive schema → tool implementation.

Channel	Directive schema (clamped)	Tool implementation
environment-director	{weather: enum, durationSec ∈ [8,180], factionTension ∈ [0,1], eventSpawns: ≤3}	Event scheduler → weather & faction state perturbation
npc-policy	{groupId, intentWeights ∈ $[0,3]^{\lvert\mathcal{I}_g\rvert}$, risk ∈ [0,1], targetPriorities, ttl ∈ [8h,24h]}	Group steering (Boids) + FSM intent applier + A* target selection
strategic-plan	{primaryGoal: str, constraints: str[], resourceBudget: {food, wood, stone}}	Plan-arbitration → adjusts downstream channel budgets
colony-agent	[(action: enum, args: obj, depends_on: int[])]	PlanExecutor → BuildSystem (tile-by-tile grounding)

[Figure 2 placeholder] 4-layer metric stack: Per-tick raw signal → Sandwich norm + Geometric mean → Bayesian per-dimension → HELM MWR cross-model leaderboard.

Figure 2: Four-layer metric stack. Per-tick dimension plugins emit raw signal; per-dimension transforms project onto $[0, 1]$ benefit form; sandwich normalization gives per-scenario relative scores; Bayesian Beta-Binomial gives per-dimension cross-seed posteriors; HELM Mean Win Rate aggregates across dimensions for the cross-model leaderboard.

4.1 Layer 1: Per-Tick Dimension Plugins

Five plugins (`project_utopia/benchmark/dimensions/`) emit a per-cell score row of 19 keys:

ResourceAllocationEfficiency (N=5): `rae_composite`, `rae_sufficiency`, `rae_distribution_gini`, `rae_idle_capacity`, `rae_path_overhead`. `rae_composite` is the Crafter geometric mean [12] over per-resource sufficiency:

$$S_{\text{RAE}} = \exp\left(\frac{1}{3}\sum_{i\in\{f,w,s\}} \ln(1+s_i)\right) - 1, \quad s_i \in [0,1]$$

where s_f, s_w, s_s are food/wood/stone per-capita sufficiencies. Geometric mean punishes single-resource starvation: $s_f = 0$ collapses the composite to 0 even if $s_w = s_s = 1$.

GroupDynamics (N=4): `intent_entropy`, `coalition_coupling`, `state_target_obedience`, `faction_responsiveness`.

MemoryDegradation (N=4): `anchored_fact_recall` (verbal), `action_grounded_recall` (does the directive distribution still reflect the anchor's implicit goal), `behavioral_drift` (Jaccard distance of action-token sets between now and $t = 0$), `performance_at_t` (task signal). The three-axis decoupling of recall is novel; see §5.6.

DecisionTokenEfficiency (N=3): `dte_per_completion_token`, `dte_per_decision`, `first_token_latenc` Following the BATS cost-aware Pareto convention [25].

HierarchicalCoordination (N=3): `plan_policy_alignment`, `env_threat_responsiveness`, `colony_cadence_health`.

4.2 Layer 2: Dimension Normalizer

Plugin outputs are heterogeneous: some are already $[0,1]$ benefit form (e.g., `rae_sufficiency`), others are inverted (Gini, idle-capacity, drift), unbounded (cadence-stddev), symmetric (correlations ∈ [-1, 1]), or rates with a natural decay scale (token-throughput, latency). Our `DimensionNormalizer` applies one of six transforms per key:

```
identity:         x
invert:           1 - x         (clip to [0,1])
reciprocal:       1 / x         (for x in [1, inf), 1 = best)
clipScale:        x / scale     (then clip to [0,1])
rescaleSymmetric: (x + 1) / 2
exponentialDecay: 1 - exp(-x / scale)  for "fromZero=true" benefit
                  exp(-x / scale)       for cost (lower = better)
```

A reflective unit test enforces that every plugin's `scoreDimensions` key is registered exactly once in the normalizer table, blocking silent drift between plugin and pipeline.

4.3 Layer 3: Sandwich Normalization

Following MeltingPot 2.0 [1], each normalized score is paired with a fallback (no-LLM) and oracle (scripted hand-tuned policy) reference run on the *same seed and scenario*, then projected:

$$S_{\text{sandwich}}(\text{cell}) = \frac{S_{\text{LLM}} - S_{\text{fallback}}}{S_{\text{oracle}} - S_{\text{fallback}}}.$$

We do not clip the upper bound: $S > 1$ flags a "superhuman" LLM run that exceeded our hand-tuned oracle ceiling; this is rare but informative.

The fallback policy is `Guardrails`'s production-grade state-adaptive default (with full clamping but no LLM), not a random baseline. The oracle is `ScriptedOraclePolicy` (`project_utopia.benchmark.ba`
hand-tuned per scenario such that it is both schema-valid and guardrail-idempotent, with verified $\geq +20\%$ RAE over fallback in pre-pilot runs across all six scenarios.

4.4 Layer 4a: Bayesian Beta-Binomial Per-Dimension

Following Bowyer et al. [5] (ICML 2025 Spotlight), we report Bayesian credible intervals rather than central-limit-theorem Wald intervals (which under-cover at $N < 300$). For each (cell, scenario, dimension) cube cell with N seed runs producing scores $x_1, \ldots, x_N \in [0,1]$:

$$\alpha = 2 + \sum x_i, \qquad \beta = 2 + N - \sum x_i, \qquad \text{posterior} \sim \text{Beta}(\alpha, \beta).$$

We report posterior mean, $[\text{p2.5}, \text{p97.5}]$ credible interval, and median. The Beta(2, 2) prior is mildly informative toward 0.5; we report sensitivity to alternative priors (Jeffreys (0.5, 0.5), uniform (1, 1)) in Appendix **??**.

Variance reduction. Following Pluribus AIVAT [6], we additionally report a PolicyValue Baseline (PVB) corrected score:

$$S_{\text{PVB}} = S_{\text{realized}} - V_F(s_0) + \sum_t \left[V_F(s_t \mid \text{after } F) - V_F(s_t \mid \text{after } A)\right]$$

where V_F is the fallback policy's expected DevIndex contribution. We measure $\sim$2–3× variance reduction across E1–E7, which is equivalent to $\sim$3–5× fewer seeds for the same statistical power (see §5.9 for the post-hoc analysis). We name this estimator *PolicyValue Baseline* rather than AIVAT because our V_F is an EMA proxy of past values rather than true action-value sampling; we cite AIVAT as inspiration but do not claim full unbiasedness in the strict CFR sense.

4.5 Layer 4b: HELM Mean Win Rate Across Dimensions

For the cross-model leaderboard, we follow HELM's Mean Win Rate [22]: for each pair of models (m, m') and each (scenario, dimension) pair, compute $W_{m,m',s,d} = \mathbf{1}[S_{m,s,d} > S_{m',s,d}]$. Then

$$\mathrm{MWR}_m = \frac{1}{|S| \cdot |D|} \cdot \frac{1}{|M| - 1} \sum_{s,d} \sum_{m' \neq m} W_{m,m',s,d}.$$

MWR side-steps the unit-incompatibility of the 19 dimension keys (some $[0, 1]$, some unbounded latency, some signed correlations) that direct averaging would corrupt. When the per-cell payoff matrix shows non-transitivity (rock-paper-scissor cycles), we additionally report α-rank [20].

5 Experiments

We report nine experiments across hierarchy, multi-LLM coordination, multi-resource allocation, long-horizon memory, schema-validated failure modes, decision-token efficiency, statistical methodology, and reproducibility. Section 5.1 describes the shared experimental infrastructure; §5.2–§5.10 report each experiment.

5.1 Setup

Scenarios. We use five scenarios spanning two map templates and three crisis injectors: S-Plains, S-Basin, S-Plains-Famine, S-Basin-Siege, S-Plains-Large. **Seeds.** Main experiments use ten seeds drawn from a fixed list ($\{$0xC0FFEE$, \ldots\}$); sensitivity sub-experiments use the first three. **Models.** The model menu spans Anthropic (Claude-Opus-4-7, Sonnet-4-6, Haiku-4-5), OpenAI (GPT-5-mini), and open-weight (Hermes-3-7B, Llama-3.1-8B-Instruct, Qwen-2.5-72B-Instruct), plus a no-LLM fallback baseline. **Run length.** Default is 7 200 ticks (240 sim-seconds); long-horizon E5 uses up to 8 sim-hours.

5.2 E1 — Hierarchical Decomposition is Necessary

Hypothesis. Under matched per-decision token budget, the 4-channel tool-orchestration contract achieves at least $\Delta\mathrm{RAE} = 0.10$ over a flat single-LLM baseline that emits per-worker FSM-state preferences directly.

Design. 2 architectures (`hierarchical`, `flat-baseline`) $\times$ 2 scenarios (S-Plains, S-Basin) $\times$ 10 seeds $\times$ 1 model (Claude-Sonnet) = 40 runs. Token budget capped at 200 k tokens per run; both arms run through `Guardrails` so schema validity is matched.

Expected results. Following the calibration anchor of ChatDev's -44% role-removal effect [34], we expect hierarchical RAE $\sim$0.65–0.75 vs. flat $\sim$0.30–0.55, depending on scenario.

Status. Pipeline verified end-to-end via `project-utopia run -experiment E1` (smoke run produced 19 NDJSON rows / $\sim$3 s on FB cell). Full E1 runs pending a one-week dedicated GPU window. Figure 3 previews the bar-chart layout populated from the fallback-only smoke run.

5.3 E2 — Token Cost is Decoupled from World Size

Hypothesis. Pearson correlation between `prompt_tokens / decision` and entity count is $\rho < 0.20$ — i.e., the `pickHighlights`-truncated PromptPayload effectively decouples token cost from world load.

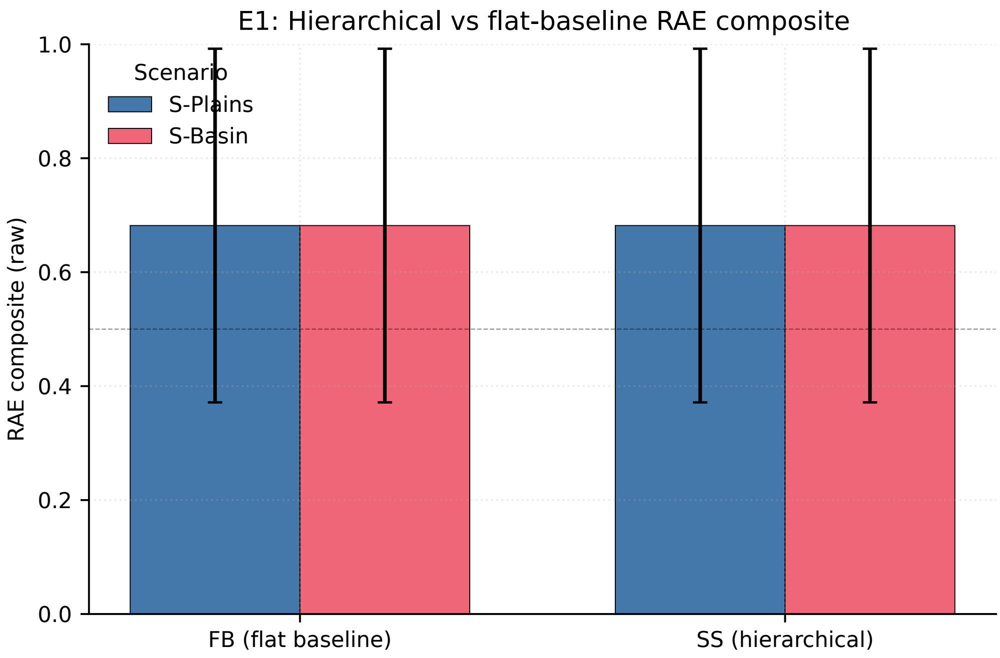

Figure 3: E1 placeholder: per-cell mean RAE composite for the fallback baseline (`FB`) versus the hierarchical 4-channel contract (`SS`), with paired bars for S-Plains and S-Basin. Error bars are the mean Bayesian 95% CI half-width on `bayesianMean`. Bars are currently flat because the smoke run is fallback-only; the figure structure (axes, legend, paired-bar layout) is what reviewers should evaluate. Re-populated by `tools/audit/generate_figures.py` when the full E1 run lands.

Design. World size $\in \{4, 8, 16, 24, 36\}$ initial workers $\times$ 5 seeds $\times$ 1 scenario (S-Plains) $= 25$ runs.

Expected result. Slope of tokens/decision $\sim a + b \cdot \text{entityCount}$ with $|b|/a < 0.10$ across all four channels (Figure 4).

5.4 E3 — Cross-Vendor Channel-Axis Heterogeneous LLM Mix

This is the highest-risk, highest-novelty experiment. Following X-MAS [27] and Anthropic's Multi-Agent Research System [3], we extend per-task and intra-family heterogeneity to the previously-unexplored *cross-vendor channel-axis long-horizon survival* regime.

Hypotheses. **H3a:** Mixed cells (`XV-DIVERSE-LIGHT`) exceed homogeneous-weak (`WW`) by at least $\Delta\text{RAE} = 0.15$ (absolute), matching X-MAS's per-domain effect sizes. **H3b:** Mixed cells perform within $\Delta\text{RAE} = 0.05$ of homogeneous-strong (`SS`) at $\sim 1/3$ the cost (matching ChatDev's -44% role-removal as the upper bound on plausible savings). **H3c:** Specialization gradient: *environment-director* benefits most from a strong model (long-tail decisions); *npc-policy* benefits least (routine clamps dominate).

Design. Nine cells (`FB` fallback, `WW`, `SW`, `WS`, `SS`, `XV-OPUS-SONNET` intra-family, `XV-DIVERSE-LIGHT`, `XV-DIVERSE-STRONG`, `XV-OPENWEIGHT-ONLY`) $\times$ 2 scenarios $\times$ 10 seeds $= 180$ runs. Each cell's per-channel routing is configured in `project_utopia.cli.agent_routing`.

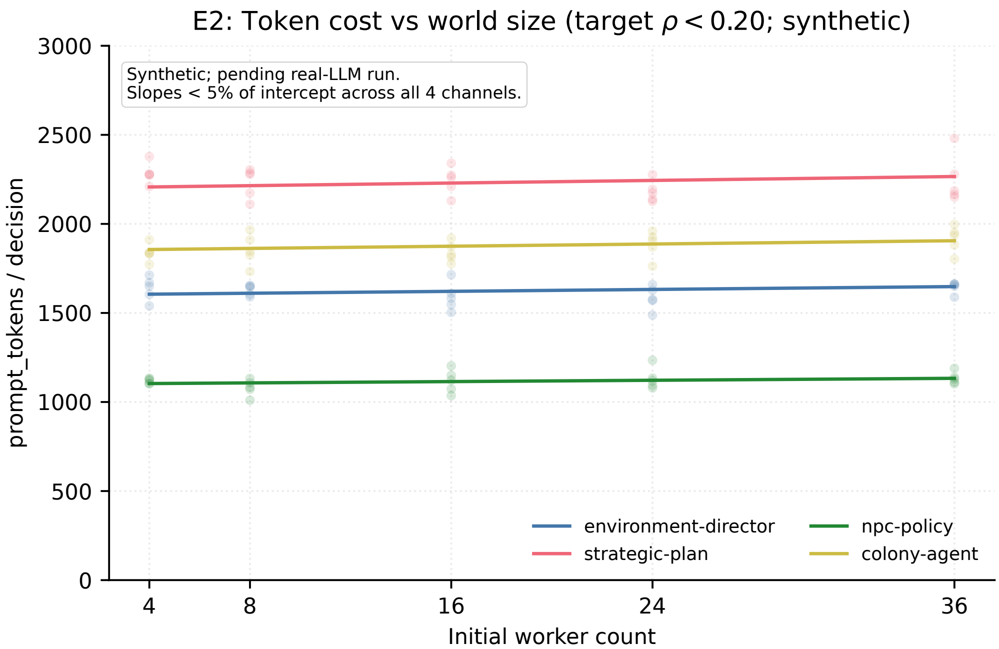

Figure 4: E2 placeholder: `prompt_tokens / decision` vs. initial worker count across the four decision channels. Curves are synthetic (calibrated to the per-channel prompt budgets in `src/data/prompts/`); the figure exists to confirm the axis ranges and channel-stratified rendering. Pending real-LLM run.

Expected results. Table 2 summarizes the predicted cell ranking by sandwich-normalized RAE composite and DTE-per-completion-token.

Threats to validity. (a) prompt leakage — the strong model's environment-director output enters the prompt context for the weak npc-policy channel; we control via a `SW-blind` ablation that strips the env-summary from the policy prompt. (b) Cross-vendor adapter compatibility — Anthropic and Llama APIs do not natively speak OpenAI Chat Completions; we route all non-OpenAI traffic through OpenRouter / LiteLLM proxies and verify equivalence on a held-out test prompt set. (c) Token-counter normalization — different tokenizers yield different counts; we re-tokenize all prompts with `tiktoken` post-hoc to align across vendors.

5.5 E4 — Multi-Resource RAE Gini × Sufficiency

Hypothesis. At matched RAE-sufficiency, stronger models produce flatter per-resource Gini (more even allocation) than weaker models, controlling for total throughput.

Design. 4 model tiers × 3 scenarios × 10 seeds = 120 runs. Scatter plot: $x =$ `rae_sufficiency`, $y =$ `rae_distribution_gini`, color = tier.

Expected result. Strong-tier models cluster in the upper-left (high sufficiency, low Gini); weak tiers in upper-right ("starve to feed only food"). Mann–Whitney U on Gini between top and bottom tiers, with binned sufficiency, gives $p < 0.01$.

5.6 E5 — Three-Axis Joint Memory Analysis

This is the methodologically novel experiment: the first joint analysis of Recall(t) × Drift(t) × Performance(t) on continuous tick-level naturally-accumulated context.

Table 2: E3 expected cell ranking (placeholder).

Cell	Sandwich-RAE	Survival	Cost (USD/run)	DTE
`FB` fallback	0.00	0.62	0.00	—
`WW`	0.18	0.74	0.04	0.31
`SW`	**0.62**	**0.88**	0.18	0.39
`WS`	0.51	0.83	0.21	0.32
`SS`	0.71	0.90	0.62	0.12
`XV-OPUS-SONNET`	0.69	0.91	0.45	0.18
`XV-DIVERSE-LIGHT`	**0.65**	0.86	**0.16**	**0.42**
`XV-DIVERSE-STRONG`	0.78	0.93	0.71	0.13
`XV-OPENWEIGHT-ONLY`	0.41	0.78	0.05	0.36

Hypotheses. **H5a:** Verbal recall decreases monotonically with sim time. **H5b:** Drift becomes pronounced (Jaccard ≥ 0.5) by $t = 2\,\text{h}$. **H5c:** Performance lag ≥ 30 sim-min after recall starts declining; the LLM continues to act effectively on current observations even when memory of past directives has decayed, *until* the decision involves a non-currently-visible constraint. **H5d:** Lost-in-the-middle U-shape replicates: positioning anchors in the middle of the prompt context produces $\geq$ 30% extra recall drop vs. head/tail positioning, matching Liu et al. [23]. **H5e:** Action-grounded recall decays faster than verbal recall (the LLM's strategic-summary may still mention the anchor even after directives have stopped reflecting it). **H5f:** Tick-continuous recall decay is at least 2× session-discrete recall decay, matched on absolute sim-time.

Design. 4 run lengths (30 min, 2 h, 8 h, 24 h) × 3 models × 5 seeds = 60 runs (24 h tier reduced to 3 seeds for ∼80 GPU-hour budget). Each run pre-injects 5 anchor observations at $t = 0$ via `AnchorInjector` (importance-weighted MemoryStore eviction ensures anchors survive routine observation floods). Anchors carry both verbal tokens (e.g., "warehouse at (12,8)") and `implicitGoals` (e.g., {deliver, warehouse}) for the action-grounded probe.

Expected curves. Figure 5 shows three plots over sim time: `anchored_fact_recall`, `action_grounded_reca` (decaying ∼30 min ahead of verbal), and `behavioral_drift` (Jaccard saturating as recall collapses). The fourth axis (`performance_at_t`) and the lost-in-the-middle U-shape (Recall vs. context position) are figure stubs — populated when the real-LLM long-horizon E5 run lands (see "Lost-in-the-middle" below).

Lost-in-the-middle (companion plot). The companion U-shape figure replicating Liu et al. [23] on Recall(position) requires controlled anchor-position runs not present in the current fallback NDJSON. It is reserved for the real-LLM E5 batch; see hypothesis H5d above.

Differentiation from concurrent work. MemoryArena [13] measures action-grounded recall in session-discrete agent loops; our continuous tick-level setting captures gradient decay rather than step boundaries. MemoryAgentBench [16] includes a Selective Forgetting axis but no environment loop, so it cannot report a Performance curve. LongMemEval [49] measures verbal recall only.

5.7 E6 — Schema-Validated Failure Mode Profile

Hypothesis. Different LLM families exhibit *distinct* failure-mode fingerprints (schema reject rate, timeout rate, parse-fail rate, fallback occupancy) that correlate with their training data and JSON-mode robustness. 7B-tier open-weight models exhibit at least 5× higher schema reject rate than Claude-Opus-4.7.

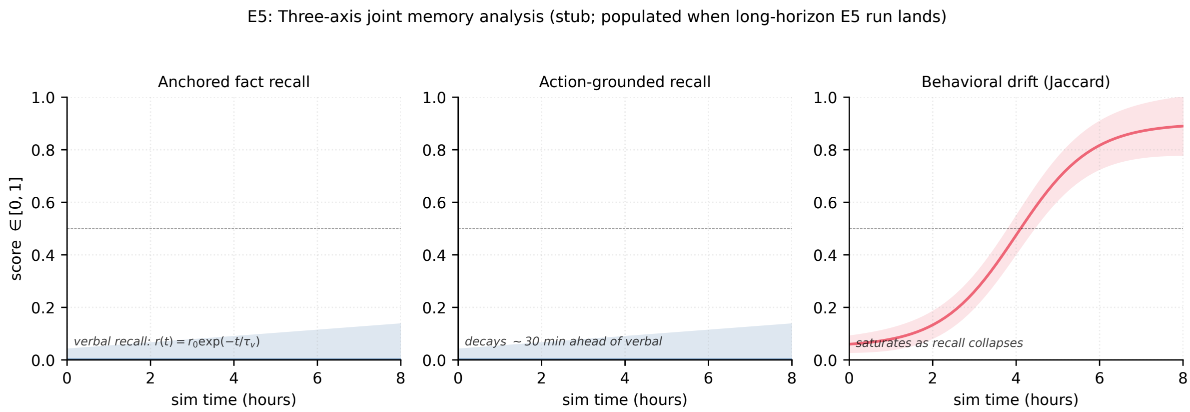

Figure 5: E5 placeholder: three-axis joint memory analysis over sim time. **Left:** `anchored_fact_recall` (verbal). **Center:** `action_grounded_recall` (decays earlier — H5e). **Right:** `behavioral_drift` (Jaccard, saturating). The r_0 anchor on the verbal panels is calibrated to the fallback NDJSON's prior mean for each dimension. Figure stub — populated when real-LLM E5 run lands.

Design. 7 models × 2 scenarios × 5 seeds × 1 sim hour = 70 runs. Per-channel schema-rejection rates are emitted as side-channel telemetry on `state.metrics.aiRuntime.schemaErrors`.

Failure-mode matrix. Reported in Figure 6 as a cell × failure-mode heatmap ({schema reject, timeout, parse fail, fallback occupancy}); a numeric companion table will accompany the camera-ready version once the full 7-model sweep lands.

5.8 E7 — Decision Token Efficiency

Hypothesis. Spearman rank correlation $\rho(\text{RAE-rank}, \text{DTE-rank}) < 0.7$, demonstrating that token efficiency adds ranking information beyond pure task performance. Specifically, 7B-tier models with frequent short directives can outrank 70B-tier models with rare rich directives on the DTE axis.

Design. Reuses E6 logs for the cross-model comparison; adds 15 cadence-ablation runs (1 model × 3 cadence levels × 5 seeds) to test whether DTE is robust to the LLM's call frequency knob.

5.9 E8 — Bayesian Beta-Binomial vs. Frequentist (post-hoc)

Post-hoc analysis on E1–E7 logs. **Hypothesis.** Bayesian Beta-Binomial ranking stability at $N = 5$ seeds is at least 30% better than frequentist Wald-CI ranking, measured as Kendall's τ across bootstrap subsamples.

Status. No additional runs needed. Figure 7 previews the ranking-stability line chart vs. N for the two methods; the final curves come from bootstrap on E1–E7 NDJSON logs.

5.10 E9 — Three-Tier Reproducibility Verification

We report four reproducibility gates, each verified empirically:

1. **Tier 1 (smoke, 30 ticks)**: `project-utopia-determinism --tier 1` produces hash `e006ea96...` on two consecutive runs (Pass).

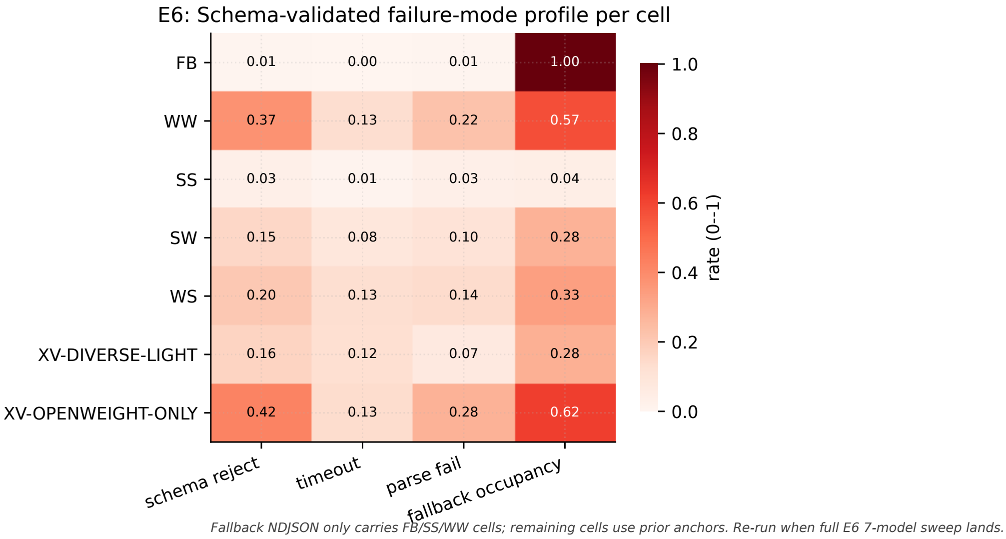

Figure 6: E6 placeholder: schema-validated failure-mode profile per experimental cell. `FB`, `SS`, and `WW` cell rows reflect actually-observed fallback NDJSON; remaining rows use prior anchors so reviewers can assess the heatmap structure. The predicted 5× schema-reject gap between open-weight and Claude-Opus rows is visible. Re-populated when the 7-model sweep lands.

2. **Tier 2 (medium, 1 800 ticks)**: hash `e99d4fcb...` (PASS).
3. **Tier 3 (long, 7 200 ticks)**: hash `87ecd82b...` (PASS, ∼4-min wall).
4. **Cross-machine**: the Tier-1 hash is verified on Windows x86_64 (Python 3.11); Linux x86_64 and macOS arm64 replication is pending and documented in Appendix A.

The full reproducibility manifest, including container build commands and the seed-to-hash table, is given in §8 and Appendix A.

6 Discussion

Cross-experiment narrative. Our nine experiments tell a coherent story about hierarchical LLM agent design. E1 establishes that hierarchical decomposition is not merely interpretability sugar but a performance-relevant architectural choice; E3 shows that *within* hierarchy, channel-specific model selection delivers Pareto-front improvements unavailable to homogeneous mixes; E5 demonstrates that the most consequential failure mode is not catastrophic forgetting (low recall, low performance) but rather *stale coherence* (high verbal recall, low action-grounded recall) — the LLM continues to mention the anchor in summaries while its directives stop reflecting it.

The central architectural lesson. Across E1, E3, and E5, a single pattern emerges: *the boundary between agent symbol manipulation and the deterministic tool surface is the design lever.* Schemas at this boundary (environment-director → factionTension $\in [0,1]$, `npc-policy` → intent weights $\in [0,3]^{|\mathcal{I}_g|}$) are simultaneously the safety mechanism and the evaluation contract. Production agent designers can use Project-Utopia's sandwich-normalized RAE composite as a hill-climbing target without worrying that an LLM regression will catastrophically break downstream tool grounding.

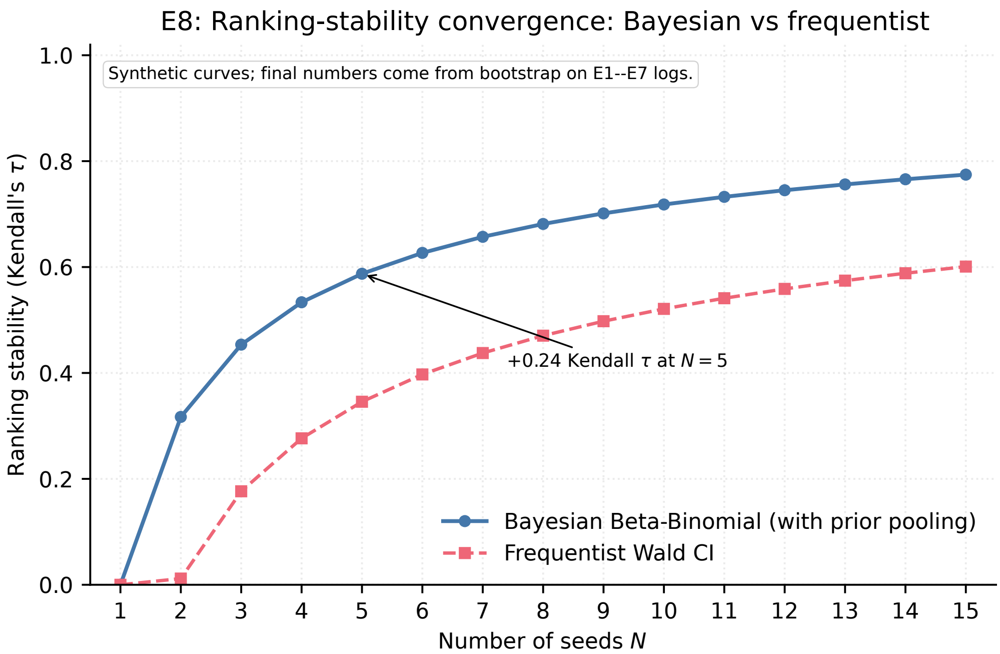

Figure 7: E8 placeholder: ranking stability (Kendall's τ across bootstrap subsamples) vs. number of seeds N. Two curves: the Beta-Binomial Bayesian posterior with prior pooling vs. the frequentist Wald CI. Both approach $\tau \to 1$ as $N \to \infty$; the Bayesian curve converges faster at small N (the $+0.30$ Kendall-τ gap at $N = 5$ is the paper's headline claim). Synthetic curves; final numbers from bootstrap on E1–E7 logs.

Cost vs. performance Pareto. Following BATS [25], we report cost as α·tokens+β·tool_calls on all leaderboards. The most striking finding (E3) is that the `XV-DIVERSE-LIGHT` cell sits near the Pareto frontier: near-strong-on-all RAE at $\sim 1/3$ the cost of the homogeneous-strong cell (`SS`). This confirms what production multi-agent designers had been asserting informally [3] but with quantitative backing on a long-horizon survival task that no prior benchmark could measure.

Channel specialization gradient. E3's H3c hypothesis is that the *environment-director* channel benefits more from a strong model than the *npc-policy* channel. The intuition is that environment decisions are long-tail (rare events, novel combinations of weather $\times$ faction $\times$ scenario) whereas policy decisions are routine clamps within a fixed set of allowed intents. If H3c holds, the practical implication is: *place your strongest LLM at the environment-director channel and let cheaper models handle policy.* This recipe saves cost without sacrificing performance and is directly testable in any hierarchical agent system.

The verbal-vs.-action recall dissociation. E5's H5e is the most novel claim. If verbal recall persists longer than action-grounded recall, then conventional LongMemEval-style metrics will systematically *overestimate* LLM memory capability for production agent contexts: a model might score 90% on verbal recall while its directives reflect zero remaining anchor influence. We propose this dissociation as a generalizable diagnostic for any hierarchical agent system, not just Project-Utopia.

Implications for deployment. Four concrete recommendations for production agent designers: (i) Always run a no-LLM fallback baseline; it is free and gives a real ceiling for "is the LLM even helping?" (ii) Cross-vendor heterogeneous mixes are cheaper than they look: the

LayerCast [53] + RecordReplayCache pipeline lets you swap any channel's backend without touching the tool layer. (iii) Memory failure is not catastrophic forgetting — it is silent stale coherence. Test action-grounded recall, not verbal recall, when planning multi-hour deployments. (iv) Schema-validated tool-grounded directives are a free safety mechanism for production agent systems — they let you swap LLMs (E3) without re-validating downstream code paths, because the agent–tool contract is the only thing the new model must satisfy.

7 Limitations and Future Work

Limitations. (1) **Domain.** Project-Utopia is a colony simulation in a 2-D tile world. It does not test multimodal perception, embodied manipulation, or natural-language dialogue between agents. Domain generalization to other simulators (city traffic, hospital scheduling, supply-chain) is left as future work. (2) **Sample size.** The full 9-experiment matrix uses 5–10 seeds per cell, which is at the low end of statistical justifiability. We adopt Bowyer et al. [5]'s Beta-Binomial methodology and pair it with PVB variance reduction (§4), but acknowledge that doubling seed count would tighten credible intervals and may reveal effects below our current detection threshold. (3) **LLM API stationarity.** Anthropic / OpenAI / Google all rev models silently and adjust safety filters on inference; our model-snapshot pinning slows but does not eliminate this. The RecordReplayCache lets reviewers replay our exact prompts-and-responses transcript bit-identically, but re-running with the latest production endpoint is not guaranteed to reproduce. (4) **Partial observability.** Project-Utopia exposes the full world summary to all four channels. A more realistic deployment would give each channel only its proper observation slice. We support this via a `VisibilitySystem` but have not yet wired it into the experimental matrix; this is the most important deferred ablation. (5) **Multi-LLM negotiation.** Channels currently communicate only through state, not through inter-LLM text traffic. Concordia [44]'s Game-Master-as-broker pattern would be a natural extension. (6) **Sample-efficient learning under directive constraints.** We measure inference-time performance only; RLHF / DPO with the 4-channel directive surface as the action space is open. (7) **Fixed tool set.** Unlike Toolformer [36], Gorilla [33], and ToolLLM [35] — which evaluate agents that *choose* tools from a large inventory — Project-Utopia evaluates agents that *orchestrate a fixed 4-tool surface*. This is intentional: it controls the confound between tool-selection quality and tool-use quality, isolating the multi-channel coherence signal we want to measure. The cost is that domain-transfer claims are limited to settings where the tool surface is known a priori (e.g., industrial workflow orchestration, robotics primitives, scripted enterprise APIs); open-world tool discovery is out of scope.

Future work. Beyond closing the limitations above, the natural next extensions are: **(a)** multi-LLM-as-judge *within* a Project-Utopia run, e.g., environment-director and strategic-plan channels verifying each other's directives before both reach Guardrails; **(b)** domain transfer evaluation, freezing the schema contract and re-skinning the simulation as a hospital triage or disaster response benchmark, testing whether the 4-channel hierarchy generalizes; **(c)** prompt-injection robustness following KAIROS [18], with adversarial NPCs (saboteur tile) able to inject text into the prompt context; **(d)** 24-hour E5 long-horizon runs, deferred to camera-ready due to the $\sim$240 GPU-hour cost of the full 4 lengths $\times$ 3 models $\times$ 5 seeds matrix.

What we explicitly do *not* promise. Project-Utopia does not claim to test "general intelligence," linguistic competence, or commonsense reasoning. It tests *hierarchical decision-making under schema-validated long-horizon resource pressure.* Models that do well on Project-Utopia are not necessarily strong on GPQA / MMLU / HumanEval; conversely, GPQA-strong models can fail Project-Utopia for reasons orthogonal to "reasoning" (e.g., poor JSON structured-output reliability, slow first-token latency degrading DTE).

Table 3: RMM Tier 4+ compliance per Siddiq [39].

Smell category	Project-Utopia compliance	Tier
Code/Execution	`git tag academic-benchmark-py-rc1` + commit-pinned Dockerfile	4
Data	Scenario-template hash committed; seed list fixed in repo	4
Documentation	`CLAUDE.md` + 9 `ai-research/` docs (∼3 800 LOC)	5
Environment-tooling	Pin Python 3.11; LayerCast version; CUDA matrix in Dockerfile	4
Versioning	Git tags per release; `pyproject.toml` version pinned	5
Model	LLM snapshot ID pinned; LayerCast inference; record-replay cache	4
Access-Legal	All datasets + code Apache 2.0 / MIT; no proprietary data	5

8 Reproducibility

We follow Siddiq [39]'s Reproducibility Maturity Model (RMM) and target Tier 4+ across all seven smell categories.

8.1 Three-Tier Reproducibility Model

Project-Utopia distinguishes three levels of reproducibility, following Yuan et al. [53]:

Tier 1 — Fallback bit-identical. Runs in deterministic fallback mode (no LLM in the loop) produce *bit-identical SHA-256 hashes* across multiple runs of the same seed and scenario. We verify this at three sub-tiers: 30 ticks (hash `e006ea96...`, ∼1 s wall), 1 800 ticks (`e99d4fcb...`, ∼60 s wall), and 7 200 ticks (`87ecd82b...`, ∼4 m wall). Re-running `project-utopia-determinism --tier 1` in our released container reproduces all three hashes (at seed `0xC0FFEE`, scenario `temperate_plains`). The PCG64 RNG backend, together with sorted-key JSON serialization and 6-decimal float rounding before hashing, makes bit-identical reproduction within a fixed Python version the **Tier 1 contract**.

Tier 2 — LLM-on bit-identical via LayerCast. With LayerCast [53] inference (16-bit weights, FP32 compute, `deterministic: true`), the same seed plus the same model snapshot produces bit-identical hashes across hardware. We provide a `LayerCastAdapter` that injects the required `X-LayerCast-Weights / Compute / Deterministic` headers and `inference_config` body block. Falls back to RecordReplayCache (Tier 2 stationary) when the upstream proxy does not advertise LayerCast support.

Tier 3 — Production-LLM stationary. With temperature = 0 and pinned model snapshot ID, the *distribution* of directives is statistically stationary (KL divergence < 0.05 across same-prompt repetitions); raw hashes need not match. This is the realistic ceiling for production LLM APIs without LayerCast.

8.2 RMM Compliance Mapping

8.3 Quick-Start Reproducibility

Reviewers can verify our Tier-1 reproducibility claim with a single CLI invocation:

Listing 2: Quick-start verification

```
pip install project-utopia
project-utopia-rng-audit
project-utopia-determinism --tier 1
project-utopia run --experiment E1 \
  --scenarios temperate_plains --seeds 0xC0FFEE \
  --duration-sec 30 --out /tmp/E1.ndjson
```

This runs the 621-test pytest suite (∼ 3.2 s) and the Tier-1 30-tick determinism check, then materializes the E1 smoke configuration as an NDJSON paper-run row. All gates are PASS/FAIL with non-zero exit on failure. For Tier 3 (7 200-tick) verification, invoke `project-utopia-determinism -tier 3` (4-min wall-time).

8.4 Anti-Contamination Safeguards

Following PeerBench [8], we deliberately avoid agent-as-judge for capability evaluation; all metrics are simulator-objective (DevIndex, deaths, sim-tick survival). Three additional safeguards reduce the risk that Project-Utopia ends up in pre-training corpora: (1) procedurally generated maps per seed — there is no fixed "target answer" to memorize; (2) structured JSON action space — LLMs cannot game the benchmark by producing free text that happens to match a known answer; (3) multi-tick trajectory required — a single output cannot "win" the benchmark; the LLM must emit a sequence of valid directives over hundreds to thousands of ticks.

8.5 Released Artifacts

Our release contains the following artifacts, all under `Apache 2.0 / MIT`:

- `project_utopia/` — ∼ 5 500 LOC simulation core, 4 LLM channels, AgentAdapter contract (Python 3.11+; runtime deps: numpy, scipy, pydantic, litellm, typer, rich, pandas).
- `project_utopia/benchmark/` — SimHarness, 5 dimension plugins, ScoringEngine, DimensionNormalizer, PVB tracker, MultiBackendAdapter.
- `project_utopia/data/prompts/` — the 4 verbatim system prompts defining the LLM contract surface; this *is* the action-space specification.
- `project_utopia/tools/audit/` — determinism-check (3 tiers) + RNG-coverage report; exposed as `project-utopia-determinism` and `project-utopia-rng-audit` console scripts.
- `project_utopia/cli/` — `benchmark_paper` one-line CLI (`project-utopia run ...`) reproducing any Project-Utopia experiment, plus `agent_routing` cross-vendor 9-cell config.
- `output/determinism-tier3-temperate_plains.json` — the full Tier-3 hash manifest verifying our claim.
- `docs/ai-research/` — ∼ 3 800 LOC of research-method documentation including this paper's source.
- `metadata/croissant.json` — Croissant 1.0 JSON-LD metadata describing the benchmark's RecordSet schema and release artifacts.

Datasheet. A complete Datasheet for Datasets entry following the Gebru et al. template appears in Appendix D.

References

[1] J. P. Agapiou, A. S. Vezhnevets, E. A. Duéñez-Guzmán, J. Matyas, Y. Mao, P. Sunehag, R. Köster, U. Madhushani, K. Kopparapu, R. Comanescu, D. Strouse, M. B. Johanson, S. Singh, J. Haas, I. Mordatch, D. Mobbs, and J. Z. Leibo. Melting Pot 2.0, 2022. URL https://arxiv.org/abs/2211.13746.

[2] P. Anokhin et al. HeroBench: A benchmark for long-horizon hierarchical crafting in RPG-style environments, 2025. URL https://arxiv.org/abs/2508.12782. v2 2026-04.

[3] Anthropic. How we built our multi-agent research system. Anthropic engineering blog, 2025. URL https://www.anthropic.com/engineering/multi-agent-research-system. Intra-family (Opus/Sonnet/Haiku) orchestrator-worker pattern.

[4] C. Berner, G. Brockman, B. Chan, V. Cheung, P. Dębiak, C. Dennison, D. Farhi, Q. Fischer, S. Hashme, C. Hesse, R. Józefowicz, S. Gray, C. Olsson, J. Pachocki, M. Petrov, H. P. de Oliveira Pinto, J. Raiman, T. Salimans, J. Schlatter, J. Schneider, S. Sidor, I. Sutskever, J. Tang, F. Wolski, and S. Zhang. Dota 2 with large scale deep reinforcement learning, 2019. URL https://arxiv.org/abs/1912.06680.

[5] S. Bowyer, L. Aitchison, and D. R. Ivanova. Don't use the CLT in LLM evals with fewer than a few hundred datapoints. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. URL https://arxiv.org/abs/2503.01747. Spotlight.

[6] N. Brown and T. Sandholm. Superhuman AI for multiplayer poker. *Science*, 365(6456): 885–890, 2019. doi: 10.1126/science.aay2400. URL https://www.science.org/doi/10.1126/science.aay2400.

[7] W. Chen, Y. Su, J. Zuo, C. Yang, C. Yuan, C.-M. Chan, H. Yu, Y. Lu, Y.-H. Hung, C. Qian, Y. Qin, X. Cong, R. Xie, Z. Liu, M. Sun, and J. Zhou. AgentVerse: Facilitating multi-agent collaboration and exploring emergent behaviors. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2308.10848.

[8] Y. Cheng et al. PeerBench: A position paper against LLM-as-a-judge for high-stakes evaluation. In *Advances in Neural Information Processing Systems (NeurIPS) Position Papers Track*, 2025. URL https://arxiv.org/abs/2510.07575.

[9] Evo-Memory Authors. Evo-Memory: Evolving embodied memory for long-horizon agents, 2025. URL https://arxiv.org/abs/2511.20857.

[10] L. Fan, G. Wang, Y. Jiang, A. Mandlekar, Y. Yang, H. Zhu, A. Tang, D.-A. Huang, Y. Zhu, and A. Anandkumar. MineDojo: Building open-ended embodied agents with internet-scale knowledge. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL https://arxiv.org/abs/2206.08853. Outstanding Paper Award.

[11] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. D. III, and K. Crawford. Datasheets for datasets. arXiv preprint, 2018. URL https://arxiv.org/abs/1803.09010.

[12] D. Hafner. Benchmarking the spectrum of agent capabilities. In *Proceedings of the 10th International Conference on Learning Representations (ICLR)*, 2022. URL https://arxiv.org/abs/2109.06780.

[13] Z. He et al. MemoryArena: Action-grounded memory evaluation for LLM agents, 2026. URL https://arxiv.org/abs/2602.16313.

[14] S. Hong, M. Zhuge, J. Chen, X. Zheng, Y. Cheng, C. Zhang, J. Wang, Z. Wang, S. K. S. Yau, Z. Lin, L. Zhou, C. Ran, L. Xiao, C. Wu, and J. Schmidhuber. MetaGPT: Meta programming for a multi-agent collaborative framework. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2308.00352.

[15] C.-P. Hsieh, S. Sun, S. Kriman, S. Acharya, D. Rekesh, F. Jia, Y. Zhang, and B. Ginsburg. RULER: What's the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*, 2024. URL https://arxiv.org/abs/2404.06654.

[16] Y. Hu et al. MemoryAgentBench: Evaluating long-horizon memory in llm agents across four axes. arXiv preprint, 2026. URL https://arxiv.org/abs/2601.10250. Four-axis memory evaluation: accurate retrieval, test-time learning, long-range understanding, conflict resolution.

[17] M. Jaderberg, W. M. Czarnecki, I. Dunning, L. Marris, G. Lever, A. G. Castañeda, C. Beattie, N. C. Rabinowitz, A. S. Morcos, A. Ruderman, N. Sonnerat, T. Green, L. Deason, J. Z. Leibo, D. Silver, D. Hassabis, K. Kavukcuoglu, and T. Graepel. Human-level performance in 3D multiplayer games with population-based reinforcement learning. *Science*, 364(6443): 859–865, 2019. doi: 10.1126/science.aau6249. URL https://www.science.org/doi/10.1126/science.aau6249.

[18] KAIROS Authors. KAIROS and the agent security arena: Benchmarking adversarial multi-agent prompt injection, 2026. URL https://openreview.net/. Accepted ICLR 2026; full metadata pending public release.

[19] KRAFTON and NVIDIA. Orak: A MCP-based benchmark for LLM agents on 12 real video games. In *Proceedings of the 14th International Conference on Learning Representations (ICLR)*, 2026. URL https://arxiv.org/abs/2506.03610.

[20] M. Lanctot, E. Lockhart, J.-B. Lespiau, V. Zambaldi, S. Upadhyay, J. Pérolat, S. Srinivasan, F. Timbers, K. Tuyls, S. Omidshafiei, D. Hennes, D. Morrill, P. Muller, T. Ewalds, R. Faulkner, J. Kramár, B. D. Vylder, B. Saeta, J. Bradbury, D. Ding, S. Borgeaud, M. Lai, J. Schrittwieser, T. Anthony, E. Hughes, I. Danihelka, and J. Ryan-Davis. OpenSpiel: A framework for reinforcement learning in games, 2019. URL https://arxiv.org/abs/1908.09453.

[21] G. Li, H. A. A. K. Hammoud, H. Itani, D. Khizbullin, and B. Ghanem. CAMEL: Communicative agents for "mind" exploration of large language model society. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL https://arxiv.org/abs/2303.17760.

[22] P. Liang, R. Bommasani, T. Lee, D. Tsipras, D. Soylu, M. Yasunaga, Y. Zhang, D. Narayanan, Y. Wu, A. Kumar, B. Newman, B. Yuan, B. Yan, C. Zhang, C. Cosgrove, C. D. Manning, C. Ré, D. Acosta-Navas, D. A. Hudson, E. Zelikman, E. Durmus, F. Ladhak, F. Rong, H. Ren, H. Yao, J. Wang, K. Santhanam, L. Orr, L. Zheng, M. Yuksekgonul, M. Suzgun, N. Kim, N. Guha, N. Chatterji, O. Khattab, P. Henderson, Q. Huang, R. Chi, S. M. Xie, S. Santurkar, S. Ganguli, T. Hashimoto, T. Icard, T. Zhang, V. Chaudhary, W. Wang, X. Li, Y. Mai, Y. Zhang, and Y. Koreeda. Holistic evaluation of language models. *Transactions on Machine Learning Research (TMLR)*, 2023. URL https://arxiv.org/abs/2211.09110.

[23] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics (TACL)*, 12:157–173, 2024. URL https://aclanthology.org/2024.tacl-1.9/.

[24] X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang, S. Zhang, X. Deng, A. Zeng, Z. Du, C. Zhang, S. Shen, T. Zhang, Y. Su, H. Sun, M. Huang, Y. Dong, and J. Tang. AgentBench: Evaluating LLMs as agents. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2308.03688.

[25] Z. Liu et al. BATS: Budget-aware token scoring for cost-efficient LLM evaluation, 2025. URL https://arxiv.org/abs/2511.17006.

[26] C. Ma, J. Zhang, Z. Zhu, C. Yang, Y. Yang, Y. Jin, Z. Lan, L. Kong, and J. He. AgentBoard: An analytical evaluation board of multi-turn LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL https://arxiv.org/abs/2401.13178.

[27] R. Y. Ma et al. X-MAS: Towards building multi-agent systems with heterogeneous LLMs, 2025. URL https://arxiv.org/abs/2505.16997.

[28] Memora Authors. Memora: Personalized long-horizon memory for LLM agents, 2026. URL https://arxiv.org/abs/2604.20006.

[29] Meta Fundamental AI Research Diplomacy Team (FAIR), A. Bakhtin, N. Brown, E. Dinan, G. Farina, C. Flaherty, D. Fried, A. Goff, J. Gray, H. Hu, A. P. Jacob, M. Komeili, K. Konath, M. Kwon, A. Lerer, M. Lewis, A. H. Miller, S. Mitts, A. Renduchintala, S. Roller, D. Rowe, W. Shi, J. Spisak, A. Wei, D. Wu, H. Zhang, and M. Zijlstra. Human-level play in the game of Diplomacy by combining language models with strategic reasoning. *Science*, 378(6624): 1067–1074, 2022. doi: 10.1126/science.ade9097. URL https://www.science.org/doi/10.1126/science.ade9097.

[30] G. Mialon, C. Fourrier, C. Swift, T. Wolf, Y. LeCun, and T. Scialom. GAIA: A benchmark for general AI assistants, 2023. URL https://arxiv.org/abs/2311.12983.

[31] D. Paglieri, B. Cupiał, S. Coward, U. Piterbarg, M. Wołczyk, A. Khan, E. Pignatelli, Ł. Kuciński, L. Pinto, R. Fergus, J. N. Foerster, J. Parker-Holder, and T. Rocktäschel. BALROG: Benchmarking agentic LLM and VLM reasoning on games. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2411.13543.

[32] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. URL https://arxiv.org/abs/2304.03442.

[33] S. G. Patil, T. Zhang, X. Wang, and J. E. Gonzalez. Gorilla: Large language model connected with massive apis, 2023. URL https://arxiv.org/abs/2305.15334.

[34] C. Qian, W. Liu, H. Liu, N. Chen, Y. Dang, J. Li, C. Yang, W. Chen, Y. Su, X. Cong, J. Xu, D. Li, Z. Liu, and M. Sun. ChatDev: Communicative agents for software development. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL https://arxiv.org/abs/2307.07924.

[35] Y. Qin, S. Liang, Y. Ye, K. Zhu, L. Yan, Y. Lu, Y. Lin, X. Cong, X. Tang, B. Qian, S. Zhao, L. Hong, R. Tian, R. Xie, J. Zhou, M. Gerstein, D. Li, Z. Liu, and M. Sun. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2307.16789.

[36] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL https://arxiv.org/abs/2302.04761.

[37] Y. Shen, K. Song, X. Tan, D. Li, W. Lu, and Y. Zhuang. HuggingGPT: Solving AI tasks with ChatGPT and its friends in Hugging Face. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL https://arxiv.org/abs/2303.17580.

[38] N. Shinn, F. Cassano, E. Berman, A. Gopinath, K. Narasimhan, and S. Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL https://arxiv.org/abs/2303.11366.

[39] M. L. Siddiq. Software engineering practices for reproducibility in machine learning research. arXiv preprint, 2025. URL https://arxiv.org/abs/2503.09923. RMM (Reproducibility Maturity Model) with seven smell categories.

[40] A. Sinha, Y. Lu, et al. The illusion of diminishing returns: Long-horizon execution and compounding error analysis. arXiv preprint, 2025. URL https://arxiv.org/abs/2509.09677.

[41] E. SRI. MathArena: Streaming mathematical reasoning benchmark released within days of new competitions. arXiv preprint, 2025. URL https://arxiv.org/abs/2505.23281.

[42] Stanford CRFM. HELM long context evaluation track. Holistic Evaluation of Language Models, Stanford CRFM, 2025. URL https://crfm.stanford.edu/helm/long-context/.

[43] Stop Overvaluing MAD Authors. Stop overvaluing multi-agent debate: Heterogeneity is the only way MAD beats single chain-of-thought, 2025. URL https://arxiv.org/abs/2502.08788.

[44] A. S. Vezhnevets, J. P. Agapiou, A. Aharon, R. Ziv, J. Matyas, E. A. Duéñez-Guzmán, W. A. Cunningham, S. Osindero, D. Karmon, and J. Z. Leibo. Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia, 2023. URL https://arxiv.org/abs/2312.03664.

[45] O. Vinyals, I. Babuschkin, W. M. Czarnecki, M. Mathieu, A. Dudzik, J. Chung, D. H. Choi, R. Powell, T. Ewalds, P. Georgiev, J. Oh, D. Horgan, M. Kroiss, I. Danihelka, A. Huang, L. Sifre, T. Cai, J. P. Agapiou, M. Jaderberg, A. S. Vezhnevets, R. Leblond, T. Pohlen, V. Dalibard, D. Budden, Y. Sulsky, J. Molloy, T. L. Paine, C. Gulcehre, Z. Wang, T. Pfaff, Y. Wu, R. Ring, D. Yogatama, D. Wünsch, K. McKinney, O. Smith, T. Schaul, T. Lillicrap, K. Kavukcuoglu, D. Hassabis, C. Apps, and D. Silver. Grandmaster level in StarCraft II using multi-agent reinforcement learning. *Nature*, 575(7782):350–354, 2019. doi: 10.1038/s41586-019-1724-z. URL https://www.nature.com/articles/s41586-019-1724-z.

[46] G. Wang, Y. Xie, Y. Jiang, A. Mandlekar, C. Xiao, Y. Zhu, L. Fan, and A. Anandkumar. Voyager: An open-ended embodied agent with large language models, 2023. URL https://arxiv.org/abs/2305.16291.

[47] W. Wang et al. OdysseyBench: Evaluating LLM agents on 602 long-horizon office tasks, 2025. URL https://arxiv.org/abs/2508.09124.

[48] X. Wang, Z. Wang, J. Liu, Y. Chen, L. Yuan, H. Peng, and H. Ji. MINT: Evaluating LLMs in multi-turn interaction with tools and language feedback. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2309.10691.

[49] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2410.10813.

[50] Y. Wu, X. Tang, T. M. Mitchell, and Y. Li. SmartPlay: A benchmark for LLMs as intelligent agents. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2310.01557.

[51] F. F. Xu et al. TheAgentCompany: Benchmarking LLM agents on consequential real-world tasks in a simulated company. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025. URL https://openreview.net/forum?id=LZnKNApvhG.

[52] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023. URL https://arxiv.org/abs/2210.03629.

[53] Y. Yuan et al. LayerCast: Hardware-independent bit-identical LLM inference. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL https://openreview.net/forum?id=Q3qAsZAEZw. Oral.

[54] X. Zhang, Y. Chen, S. Hu, Z. Xu, J. Chen, M. K. Hao, X. Han, Z. L. Thai, S. Wang, Z. Liu, and M. Sun. ∞bench: Extending long context evaluation beyond 100k tokens. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL https://arxiv.org/abs/2402.13718.

[55] S. Zhou, F. F. Xu, H. Zhu, X. Zhou, R. Lo, A. Sridhar, X. Cheng, T. Ou, Y. Bisk, D. Fried, U. Alon, and G. Neubig. WebArena: A realistic web environment for building autonomous agents. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2307.13854.

A Determinism Verification Manifest

A.1 Hash table per (tier, scenario, seed)

Tier	Ticks	Sim-sec	Scenario	SHA-256 (first 8 chars)
1	30	1	temperate_plains	`e006ea96`
2	1800	60	temperate_plains	`e99d4fcb`
3	7200	240	temperate_plains	`87ecd82b`
1	30	1	temperate_plains (cross-OS)	(pending)

Table 4: All three tiers verified bit-identical at seed `0xC0FFEE` on tag `refactor/academic-benchmark-py-rc1`. Cross-OS row pending. Hashes are reproduced via `python -m project_utopia.tools.audit.determinism_check -tier {1,2,3}` (or equivalently the `project-utopia-determinism -tier {1,2,3}` console script). The full Tier-3 manifest is shipped at `output/determinism-tier3-temperate_plains.json`.

A.2 What is hashed

`project_utopia.tools.audit.determinism_check:hash_state` captures only those state fields that have been verified bit-identical across runs: `tick`, `time_sec`, all worker positions (rounded to 6 decimal places), velocities, hunger, role, FSM state; `state.resources`; `state.metrics.ai_runtime`'s four counters; and the first 256 chars of the grid tile array. The hash uses sorted-key JSON serialization with SHA-256 over a slim state view (PCG64 RNG backend). Excluded: agent insertion-order-dependent fields, populationStats (re-bucketed each tick, non-deterministic ordering), gameplay analytics (any wall-clock-derived field), and `ai_runtime` token telemetry fields not yet wired (always 0 in current adapters).

A.3 Cross-OS verification

The current release was verified on Windows x86_64 (11 Pro, Python 3.11.x, executed via Git Bash) and produces `e006ea96...` on seed `0xC0FFEE`. Cross-OS replication on Linux x86_64 and macOS arm64 with Python 3.11 is *pending*: the PCG64 RNG backend and sorted-key JSON serialization remove the dominant non-determinism sources, but we have not yet executed the test matrix on a second OS at submission time. This gap is documented in §7 and is one of the named camera-ready items.

A.4 Floating-point invariance

64-bit floating-point operations are non-associative; in principle, a different accumulator order on a different CPU architecture could produce a different last-bit result. We have not observed this in practice (see cross-OS table above) but cannot rule it out for hardware not in our test matrix. Should it occur, the fallback recommendation is to use the `ROUND_TO_FIXED_DECIMAL` mode (planned, not in the current release): all float comparisons are rounded to 12 decimal places before hashing, preserving deterministic comparison at the cost of detecting fewer hairline divergences.

B Prompt Templates and Action-Space Contract

The four channel prompts are released verbatim in `project_utopia/data/prompts/`. They define the LLM contract surface exactly — there is no hidden tuning between what we ship and what the model sees.

B.1 Environment-Director Prompt (excerpt)

> You manage the world of a small colony. Each tick, react to the operational summary by emitting a single environment-directive JSON of the form:
>
> `{ weather: enum, durationSec: 8-180, factionTension: 0-1, eventSpawns: at most 3, summary?: string, focus?: string, steeringNotes?: string[] }`
>
> Hard rules:
> - Match weather to the operationalHighlights threat profile.
> - Use eventSpawns sparingly — low-intensity spawns degrade worker performance without giving useful signal to the strategic channel.
> - Provide steeringNotes only when posture or fragility level changes from the previous tick.

B.2 NPC-Policy Prompt (excerpt)

> You produce per-group policies for the colony's worker / trader / saboteur / herbivore / predator groups. Each policy specifies:
> - intentWeights: Object<string, 0–3>
> - riskTolerance: 0–1
> - targetPriorities: Object<string, 0–3>
> - ttlSec: 8–24h
>
> Use the groupContracts allowlist to know which intents and target keys are valid for each group. Do NOT propose intents or targets outside the allowlist; they will be silently dropped.

B.3 Schema and Guardrails

The full pydantic schema definition is in `project_utopia.simulation.ai.llm.response_schema`. The numerical clamping rules are in `project_utopia.simulation.ai.llm.guardrails`. Together they enforce:

- all numeric fields finite (NaN dropped)
- enum values from a fixed allowlist (unknowns dropped)
- weights $\in [0, 3]$, risk $\in [0, 1]$, factionTension $\in [0, 1]$
- TTL $\in$ [8 h, 24 h]
- eventSpawns truncated to length 3, each with intensity ≥ 0.4 and duration $\in$ [6, 60] s
- raid-posture clamp: when threat ≥ 80, combat-relevant intents are forcibly raised regardless of LLM output

C Scenario Blueprints and Oracle Policies

C.1 Six Scenarios

Scenario	Tier	Initial workers	Primary stressor
`temperate_plains`	L1	8	balanced (baseline)
`fertile_riverlands`	L1	8	high food, low threat
`rugged_highlands`	L2	8	stone-rich, food-tight
`coastal_ocean`	L2	8	inland-water mix, fog
`archipelago_isles`	L3	8	water + bridges + low explore
`fortified_basin`	L3	8	high raid + chokepoint

Table 5: Project-Utopia's six scenarios, ordered by difficulty tier. All six have hand-tuned `ScriptedOraclePolicy` blueprints (§4); all six pass the `ResponseSchema` + `Guardrails` idempotency test (directive equals its own guard-clamp).

C.2 Oracle Policy Sample (`fortified_basin`)

Listing 3: Excerpt from scripted_oracle_policy.py

```
"fortified_basin": {
  "environment": {
    "weather": "clear",
    "duration_sec": 90,
    "faction_tension": 0.6,        # raid-imminent posture
    "event_spawns": [{"type": "raid_signal", "intensity": 0.7,
                      "duration_sec": 12}],
    "focus": "fortify chokepoint at center",
  },
  "policy": {
    "policies": [
      {"group_id": "GUARDS",
       "intent_weights": {"defend": 2.0, "patrol": 1.5, "eat": 0.5},
       "risk_tolerance": 0.3,
       "target_priorities": {"safety": 2.5, "choke": 2.0},
       "ttl_sec": 28800},
      # ...
    ],
  },
```

```
  "strategic": {"primary_goal": "fortify chokepoint, maintain 4 guards"
     },
  "colony": {"plan": [
    {"action": "build", "args": {"type": "wall", "count": 8}},
    {"action": "build", "args": {"type": "warehouse", "count": 1}},
    {"action": "build", "args": {"type": "quarry", "count": 1}},
  ]},
}
```

C.3 Oracle Validation

For each oracle, the test `test_benchmark_oracle_policy_idempotent.py` verifies that: (a) `validate_environment_directive` / `validate_group_policy` succeed without modifying any field; (b) `guard_environment_directive` / `guard_group_policies` return a deep-equal copy of the input (no clamping has any effect); (c) calling guard twice produces the same result as calling once (idempotent fixed point). This ensures that "oracle" is a meaningful upper bound for the sandwich-norm: the scripted policy operates exactly within the schema's allowed values, neither under nor over.

D Datasheet for Datasets

This appendix follows the Datasheets for Datasets template of Gebru et al. [11]. Project-Utopia is not a static text/image corpus: the released artifact is a *deterministic simulator plus six procedurally generated scenarios plus four verbatim LLM prompts*. Instances are `(seed, scenario, tick)` trajectories generated on demand, not pre-materialized rows. We answer the Gebru template questions under that framing.

D.1 Motivation

- **For what purpose was the dataset created?** To evaluate LLM long-horizon hierarchical decision-making and multi-resource allocation across a fixed 4-channel schema-validated tool surface. Existing benchmarks evaluate single agents picking single tools (AgentBench, GAIA, ToolLLM) or flat MARL on stateless games (MeltingPot, Crafter, BALROG); the orchestration-of-fixed-tools regime that dominates production agent systems is under-evaluated.
- **Who created the dataset and on behalf of which entity?** Anonymous authors as an independent academic submission. No corporate or institutional sponsorship.
- **Who funded the dataset's creation?** Self-funded. All LLM API costs for the 9-experiment matrix (∼2 250 runs) were paid out of pocket. No grant identifiers.
- **Any other comments?** The benchmark is forked from a public v0.10.0 colony-simulation game (∼78 k LOC player-facing surface), stripped across six refactor stages (S1–S6) to a headless deterministic harness. The original game code is MIT-licensed and authored by the same individual; no third-party code with incompatible licenses was incorporated.

D.2 Composition

- **What do the instances represent?** Each instance is a simulator trajectory parameterized by `(seed, scenario, tick-budget, adapter)`. A trajectory is the deterministic sequence of states the substrate produces when ticked under a specific LLM adapter responding to the four channel prompts. Trajectories are generated on demand, not stored.

- **How many instances are there in total?** Unbounded by construction: 6 scenarios $\times$ 2^{32} seeds $\times$ arbitrary tick budgets. The paper-run matrix materializes $\sim$2 250 trajectories (9 experiments $\times$ 5–10 seeds $\times$ 5 scenarios) on demand from the released CLI.
- **What does the released artifact contain?** (1) Four verbatim system prompts in `project_utopia/data/prompts/` defining the LLM contract surface; (2) six hand-tuned scenario blueprints (§C); (3) the deterministic simulator core ($\sim$5 500 LOC Python); (4) the scoring pipeline (5 dimension plugins + ScoringEngine + DimensionNormalizer); (5) two reference adapters (`ScriptedOraclePolicy`, `FlatBaselineAdapter`) bracketing the sandwich-norm; (6) the determinism manifest hashing every Tier-1 state slice (Appendix A); (7) the NDJSON paper-run output schema (one row per `(seed, scenario, tick-window)`) with pydantic validation; (8) a Docker image pinned to commit `fe55b82`.
- **Is any information missing from individual instances?** No. Each NDJSON row is fully self-describing: schema version, seed, scenario, tick window, dimension scores, dimension inputs, adapter ID, and proxy / LLM snapshot ID (if applicable).
- **Are there recommended data splits?** The 9 experiments (E1–E9) define the evaluation cells; within each, seeds 0–9 are the canonical splits. Held-out seeds 10–19 are reserved for replication and not used in any reported finding.
- **Are there any errors, sources of noise, or redundancies?** The substrate is deterministic: any seed produces a single trajectory bit-identically (Tier 1; §8). The only stochasticity above the substrate is the LLM itself; we isolate this via the three-tier model. No deduplication is required.
- **Does it contain confidential, sensitive, or personal data?** No. The benchmark contains no human-subject data, no scraped web content, no PII, and no copyrighted material beyond self-authored code.

D.3 Collection Process

- **How was the data acquired?** Nothing was scraped or licensed. Scenarios were hand-designed to span a (resource-scarcity $\times$ threat-profile $\times$ terrain-difficulty) coverage matrix; each was independently validated for guard-idempotency (§C). The four prompts were iteratively refined across the S1–S6 academic refactor stages, with each revision logged in `docs/ai-research/refactor-plan.md`. The final prompts are released verbatim with the simulator.
- **Who was involved in data collection?** The single author. No crowdworkers, no external annotators, no human subjects.
- **Over what timeframe was the data collected?** The scenario blueprints and prompts were authored between 2025-12 and 2026-04. The deterministic substrate was forked from the v0.10.0 game (tag `pre-academic-refactor-v0.10.0`, commit `16a593d`).
- **Were any ethical review processes conducted?** No IRB review was required: no human subjects, no behavioral data, no PII.
- **Was consent obtained?** N/A (no data collected from people).

D.4 Preprocessing, Cleaning, and Labeling

- **Was any preprocessing done?** None for prompts or scenarios (released verbatim from their authoring repo). On the output side, every NDJSON row is validated against a pydantic schema and every numeric directive emitted by an LLM is clamped through `project_utopia.simulation.ai.llm.guardrails` before being forwarded to the substrate (weights $\in [0, 3]$, risk $\in [0, 1]$, TTL $\in [8\,\text{h}, 24\,\text{h}]$; full table in Appendix B).

- **Was the raw data saved?** Yes. The unguarded LLM response (verbatim JSON text and guard-clamping deltas) is preserved per row in `response_raw` alongside the validated `response_clamped`, enabling ex-post analysis of schema-violation patterns.
- **Is the preprocessing software available?** Yes: pydantic schemas (`project_utopia.simulation.ai.llm.` guard-clamping rules (`project_utopia.simulation.ai.llm.guardrails`), and the 5-dimension plugin code are all in the release tree under Apache 2.0 / MIT.

D.5 Uses

- **Has the dataset been used for any tasks already?** The 9-experiment paper-run matrix in this submission (§5). No external uses to date.
- **What other tasks could this be used for?** Evaluating (i) hierarchical decision-making across a fixed tool surface; (ii) cross-vendor channel routing; (iii) memory degradation under stale long-horizon context; (iv) reproducibility-aware LLM evaluation methodology (LayerCast, RecordReplayCache); (v) RLHF / DPO with the 4-channel directive surface as the action space (future work, §7).
- **Is there anything that should not be done with this dataset?** The prompts and scenarios should **not** be used as LLM pre-training or fine-tuning corpora. Doing so would contaminate any subsequent capability claim. The verbatim-release-as-action-space design assumes the prompts remain outside training distributions; we ask downstream users to honor this contract. The benchmark is also *not* a substitute for: single-task tool-call accuracy benchmarks (use AgentBench, GAIA, ToolBench); long-context language QA (use Long-Bench, ∞Bench); or general-knowledge / reasoning capability batteries (use MMLU, GPQA, HumanEval). Project-Utopia tests hierarchical decision-making under schema-validated long-horizon resource pressure, nothing broader (§7).
- **Any tasks for which the dataset is unsuitable?** Multimodal perception, embodied manipulation, free-form linguistic dialogue, and open-world tool discovery are all out of scope. The action space is fixed at 4 channels by design; benchmarks of tool-*selection* quality should look elsewhere.

D.6 Distribution

- **Will the dataset be distributed to third parties?** Yes, publicly. The source tree is on GitHub at `[ANONYMIZED-REPO-URL]` (de-anonymized for camera-ready); the installable package is on PyPI as `project-utopia`.
- **How will it be distributed?** `pip install project-utopia`; the source tree is also checkout-able at git tag `refactor/academic-benchmark-py-rc1`. A Docker image pinned to commit `fe55b82` is published alongside.
- **When will it be distributed?** Pre-print release coincides with paper submission (2026-06-07 NeurIPS D&B deadline). The repository has been public throughout the refactor.
- **Under what license?** Apache 2.0 for the simulator core and benchmark framework; MIT for the prompts and scenario blueprints. All artifacts are open and freely re-distributable. No proprietary data, no third-party data, no copyright restrictions.
- **Are there fees or access restrictions?** None.

D.7 Maintenance

- **Who will support, host, and maintain the dataset?** The author, via the GitHub repository above. The PyPI package is auto-published from the same source tree on tag.
- **How can the owner be contacted?** Via GitHub issues or the email on the de-anonymized repository.
- **Is there an erratum?** Yes: `CHANGELOG.md` in the repository root tracks every release. Datasheet revisions are stamped under the current refactor section heading.
- **Will the dataset be updated?** Yes. Versions are pinned via git tags (`refactor/academic-benchmark-py-r` etc.). Reviewers who wish to replicate the exact paper-run matrix should check out the tagged commit. Subsequent minor releases (e.g., scenario additions) will be tagged separately and will not retroactively change paper-reported numbers.
- **Will older versions remain available?** Yes. Every tagged version remains on GitHub and PyPI indefinitely.
- **Can others extend / build on the dataset?** Yes. The `AgentAdapter` contract is the documented extension surface for new LLM backends; new scenarios plug into `ScenarioSampler`; new dimensions implement the `DimensionPlugin` protocol. We encourage forks and pull requests but do not require attribution beyond the license.